Cambridge International AS & A Level

# BIOLOGY (9700) P1

TOPIC WISE QUESTIONS + ANSWERS | COMPLETE SYLLABUS



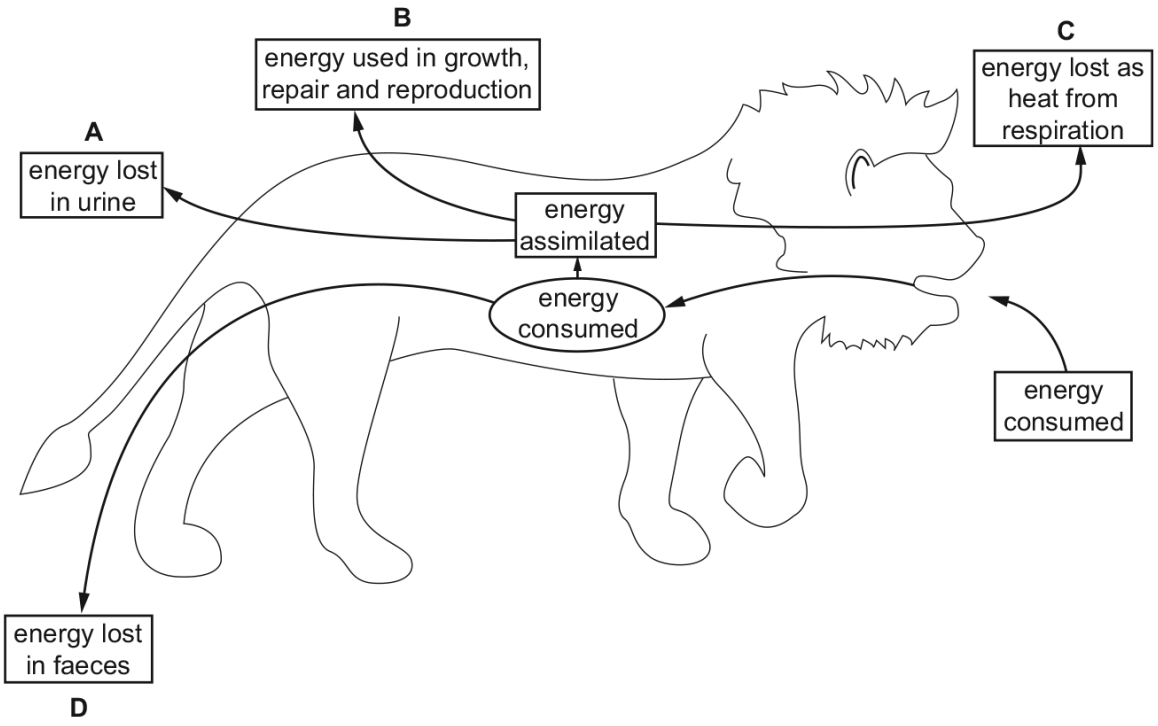
# Chapter 12

# Energy and respiration

## 12.1 Energy

1602. 9700_s15_qp_12 Q: 40

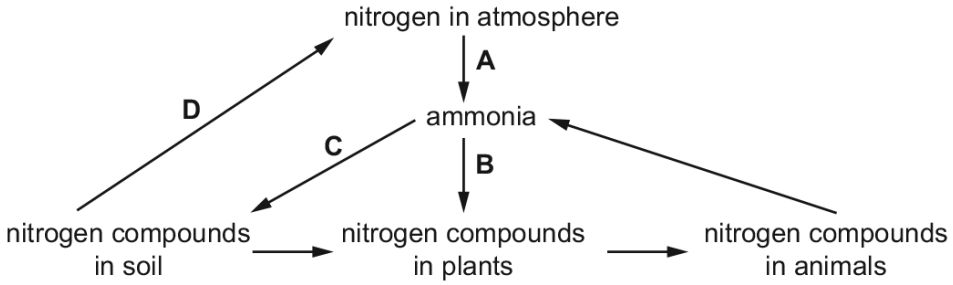
Of the prey consumed by the lion, into which box does the largest amount of energy go?



1603. 9700_s15_qp_13 Q: 38

The diagram represents part of the nitrogen cycle.

Which process is carried out by nitrifying bacteria?



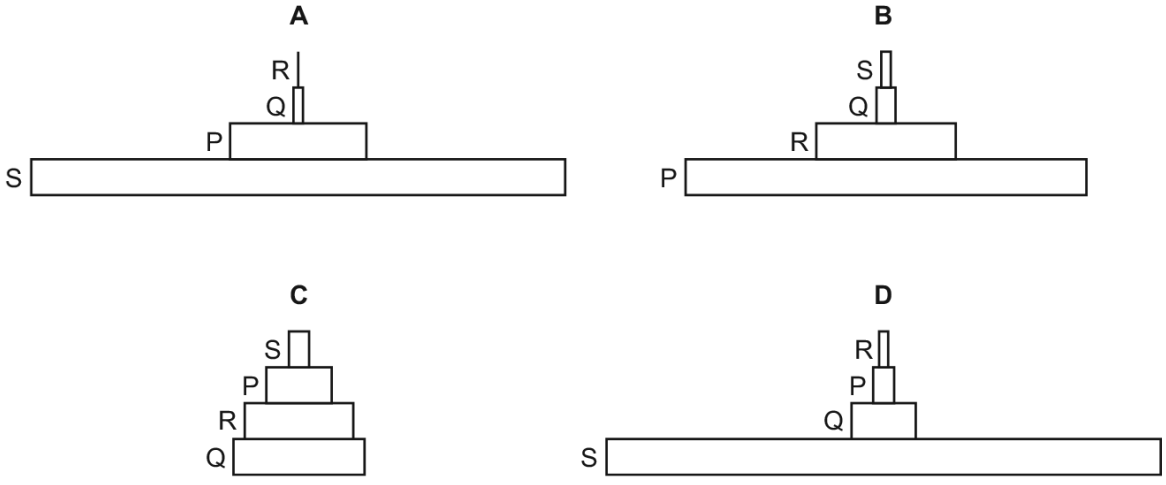

---

1604. 9700_s15_qp_13 Q: 40

The table shows the results of a field study of four species in a food chain in an area of woodland.

| species | number of individuals | biomass of one individual / arbitrary units | energy value per unit mass / arbitrary units |
|---|---|---|---|
| P | 10 000 | 0.100 | 1.0 |
| Q | 5 | 10.000 | 2.0 |
| R | 500 | 0.002 | 1.8 |
| S | 3 | 300 000.000 | 0.5 |

Which is the correct pyramid of energy from these data?




---

## 12.2 Respiration

1605. 9700_w15_qp_12 Q: 40

Anaerobic bacteria are found in large numbers in waterlogged soils.

Which effect does this have on soil fertility and why?

| | soil fertility | reason |
|---|---|---|
| **A** | decreased | bacteria convert nitrate to ammonia |
| **B** | decreased | bacteria convert nitrate to nitrogen gas |
| **C** | increased | bacteria cause decomposition |
| **D** | increased | bacteria cause nitrogen fixation |